

NIRUPAM MONDAL

West Bengal, India - 743127 | +91 7003419367 | Nirupam.mondal.me@gmail.com |
LinkedIn: Nirupam Mondal

Professional Summary

Mechanical Design Engineer with 5+ years of experience in aerospace design, electrical and mechanical system installation, digital mock-up integration, and configuration checks for Airbus commercial aircraft and helicopters. Proven expertise in CATIA V5, PLM tools (Enovia VPM, Windchill, TREND, TAKSY, COBOT Suite), and manufacturing documentation (BOM preparation, FAI sheets, Condition of Supply). Proficient in wire harness design and installation, clash clearance analysis, and volumetric studies for Airbus programs including A220, A320 family, A340, A350, H160, and H175. Skilled at collaborating in cross-functional engineering teams, interpreting technical documentation, and ensuring compliance with TRS, TDD, and SIDP requirements.

Skills

Design & CAD Tools: CATIA V5 (Surface Design, Electrical Harness Assembly, Assembly Design, DMU Navigator), CATIA Composer, CATIA CADAM Drafting, AutoCAD

Manufacturing & Documentation: Bill of Materials (BOM) Preparation, Condition of Supply Documentation, First Article Inspection (FAI) Sheets, Process Planning, Technical Review Meetings, Problem Solving Sheet (PSS) Analysis

Software & Productivity: SAP (PP Module), Microsoft Excel, Microsoft Word, Microsoft PowerPoint, Adobe Photoshop

PLM & Data Management: Enovia VPM, VPM Web, Windchill, TREND, Context Builder, ACDC/ADB, EBAM, TAKSY, CLDB, AIRINA, Skywise (FIN-Ref), APS-NG, ICC-ACP, ICC-CADB, COBOT Suite, CCD, CCD Browser, E-Check Tool, HMV, De2mat

Aerospace Programs & Systems: Airbus A220, A320, A320F, A320 TOW ASA Project, A340, A350, A350 Freighter, H160, H175; Electrical System Installation (ESI), Mechanical System Installation (MSI), Configuration Item (CI) Check, Digital Mock-Up (DMU) Integration, Clash Clearance Analysis, Volumetric Analysis, Sealant CCD Modifications, Wire Harness Design & Installation

Professional Experience

Design engineer

07/2023 to Current

Capgemini Engineering – Bengaluru, Karnataka, India

Electrical System Installation (ESI) CI Check – Airbus A320 (ATA 928 & 929)

- Executed 150+ Configuration Item (CI) checks to confirm correct implementation of design changes per Technical Repercussion Sheets (TRS).
- Cross-verified updates across 10+ PLM tools to ensure accuracy and compliance with Airbus engineering standards.

Airbus A320/A340 Design Modifications

- Modified 20+ 2D drawings for Airbus A340 (sections 16, 18, 19) to implement new sealant profiles.
- Revised A320 design sheets under the TOW ASA project, integrating updated 80-T notes and corresponding TAKSY mask changes.

Mechanical design engineer

03/2022 to 06/2023

Alten – Bangalore, India

MSI & ESI QC2 DMU Activities – Airbus A350, A350F, A320

- Performed 80+ Digital Mock-Up integration (DMUi) activities for A350, A350F, and A320, completing QC1 and QC2 checks in line with SIDP documentation for ATA92, ATA53, ATA38, ATA21, and ATA25.
- Verified harness routing and associated structures for clearance, bend radius, slack, and sagging, ensuring 100% compliance with TDD and SIDP standards.

Airbus Helicopters H175 & H160 – Clash Clearance Analysis

- Analyzed 50+ Enterprise Change Proposals (ECP) and Enterprise Change Requests (ECR) for H160 and H175.

- Verified VCIs, reviewed product structures, and prepared follow-up sheets to track and resolve clearance issues within defined Airbus timelines.

A350 Freighter – Volumetric Analysis

- Created positive volume models and subtracted surrounding structures to calculate rapid decompression volumes for 2 major fuselage sections (S13 and S15).
- Computed effective surface areas using Boolean operations in CATIA V5 to support structural safety requirements.

A320 Harness DMU Design

- Installed 200+ wire harness assemblies and supporting structures such as brackets and clamps using the Electrical Harness Assembly module in CATIA V5 and VPM.
- Created product structures in CATIA V5 per Airbus nomenclature, validating design correctness and completeness as per SIDP and TDD documentation.
- Optimized harness installations to meet bend radius, slack, and segregation requirements from surrounding environmental structures.

Methods Engineer

09/2018 to 11/2020

Tech Mahindra Pvt Ltd. – Bangalore, India

- Created 50+ visual work instructions for the Airbus A220 program using CATIA Composer, improving assembly accuracy.
- Installed 100+ structural, systems, and electrical harness components, including wire routing, clamps, connectors, and serialized parts.
- Prepared dozens of Bills of Materials (BOM) for aircraft assemblies and defined them in SAP.
- Produced 20+ Condition of Supply documents for machined and sheet metal parts in CATIA V5 based on customer input.
- Developed 15+ First Article Inspection (FAI) sheets and coordinated FAI and first-batch production activities.
- Led 10+ process planning and technical review meetings to ensure on-time delivery per customer specifications.
- Investigated and resolved 20+ escalated issues using Problem Solving Sheets (PSS), identifying root causes to prevent recurrence.

Education

Bachelor of Technology: Mechanical Engineering
JIS College of Engineering - West Bengal, India

09/2017

Languages

English

Bilingual or Proficient (C2)

German

Intermediate (B1)